AMAZON INTERVIEW EXPERIENCE

Complete Interview Experience from a Placed Student

RESOURCE	DETAIL
Type	Interview Experience / Preparation Guide
Category	Campus Placement • SDE
Format	Round-by-round interview walkthrough
Coverage	Online Assessment • DSA • System Design • Behavioral • HR
Language	English

Important note: This document is a structured, representative Amazon-style interview experience based on the supplied outline. It should not be treated as a verified account of a specific person's real interview.

Purpose: Use the round structure and question types as a preparation roadmap. Actual Amazon interviews vary by role, location, level, team and hiring process.

CANDIDATE PROFILE

Name	Rahul Gupta
Target Role	SDE-1 / Software Development Engineer
College	IIT Bombay
Batch	2024
Preparation Focus	DSA, System Design, CS Fundamentals, Leadership Principles

Preparation philosophy: Focus on explaining your reasoning, writing clean code, testing edge cases, discussing complexity, and connecting behavioral answers to concrete examples.

INTERVIEW ROUND OVERVIEW

Round	Duration	Main focus	Difficulty
1. Online Assessment	90 min	2 coding + 1 design + 15 CS MCQs	Medium
2. Technical Round 1	60 min	LLD/system design + DSA + trees	Hard
3. Technical Round 2	60 min	System design + binary search + sliding window	Hard
4. HR / Behavioral	30 min	Leadership Principles + behavioral + role fit	Easy–Medium

What interviewers evaluate: problem-solving approach, coding correctness, complexity analysis, communication, design trade-offs, ownership and behavioral judgment.

ROUND 1 — ONLINE ASSESSMENT

Duration: 90 minutes

- **Coding Problem 1 — Kth Largest Element:** Explain sorting, heap and Quickselect approaches. Be ready to discuss why a heap may be preferable when k is small.
- **Coding Problem 2 — LRU Cache:** Explain the need for O(1) get/put operations using a hash map plus doubly linked list.
- **System Design — Library Management System:** Identify entities, relationships, operations, APIs and basic class responsibilities before coding.
- **CS Fundamentals — 15 MCQs:** Revise OS processes/threads, memory, DBMS joins/indexes/normalization, networking basics and HTTP.

Suggested strategy: Do not spend the entire assessment on one difficult coding problem. Secure easier points first, then return to harder sections.

CODING DEEP DIVE — KTH LARGEST ELEMENT

Problem: Given an unsorted array, find the kth largest element.

Approach	Idea	Typical complexity
Sorting	Sort descending/ascending and access the kth position	O(n log n)
Min-heap of size k	Keep only the k largest values seen so far	O(n log k)
Quickselect	Partition around a pivot and recurse only into the relevant side	Average O(n), worst $O(n^2)$

Follow-up questions: What if the array contains duplicates? What if k is close to n? Can you implement Quickselect? What are its worst-case guarantees?

CODING DEEP DIVE — LRU CACHE

Goal: Design a cache with O(1) average-time get and put operations while evicting the least recently used item.

- **Hash map:** Maps a key to its node so the cache can locate an item in O(1) average time.
- **Doubly linked list:** Maintains recency order. The most recently used item is near the front; the least recently used item is near the back.
- **Get:** Look up the node, move it to the front, and return its value.
- **Put:** Update and move an existing node, or create a node; if capacity is exceeded, remove the least recently used node.

Key design lesson: When an interviewer asks for strict O(1) cache operations, one data structure is usually not enough; combine complementary structures to satisfy all required operations.

SYSTEM DESIGN — LIBRARY MANAGEMENT SYSTEM

Step 1 — Clarify requirements

- Who can use the system: members, librarians, administrators?
- Can a member borrow multiple books or copies?
- Do we need reservations, fines, notifications and search?
- Is the interview asking for low-level class design or a distributed system?

Step 2 — Core entities

Entity	Typical responsibility
Book	Book metadata such as ISBN, title, author, category
BookCopy	Physical/available copy of a book
Member	User who can borrow/reserve books
Librarian	Manages inventory and lending operations
Loan	Tracks checkout, due date, return and fine
Reservation	Tracks a member's reservation request

Good interview behavior: State assumptions before drawing classes. Then explain relationships, important methods, constraints and edge cases rather than trying to design every detail.

ROUND 2 — TECHNICAL ROUND 1

Question	What to demonstrate
Design a Library Management System	Requirements, entities, relationships, APIs/classes, trade-offs
Implement an LRU Cache	Hash map + doubly linked list, O(1) operations
Find Kth Largest Element	Heap vs Quickselect, complexity and edge cases
Tree Traversals	Inorder, preorder, postorder, iterative vs recursive

Technical-round rule: Do not immediately write code. Spend the first few minutes clarifying inputs, constraints and expected complexity. Explain your approach before implementing it.

ROUND 3 — TECHNICAL ROUND 2

Question	Preparation points
Design a URL Shortener	API design, ID generation, database/cache, redirects, scale and abuse cor
Binary Search Variations	First/last occurrence, lower bound, upper bound, rotated arrays, answer-sp
Sliding Window Problems	Fixed vs variable windows, frequency maps, two pointers

URL SHORTENER — HIGH-LEVEL APPROACH

- **POST /urls:** accept a long URL, validate it, generate a unique short identifier and persist the mapping.
- **GET /{code}:** retrieve the long URL from cache/database and return a redirect.
- **Storage:** use a database for durable mappings and a cache for frequently accessed codes.
- **Scale:** add load balancers and horizontally scalable application instances.
- **Reliability:** consider replication, monitoring, rate limiting and graceful failure.

ROUND 4 — HR / BEHAVIORAL

Duration: approximately 30 minutes

Question	Strong answer direction
Why Amazon?	Connect the role to engineering growth, customer impact, scale and the op
Tell me about a difficult challenge.	Use STAR: situation, task, action, result. Explain your personal contributio
Tell me about a conflict.	Show listening, evidence-based discussion, compromise and outcome.
Tell me about a failure.	Own your part, explain what you learned and what you changed afterward.
Describe leadership.	Give a concrete example of taking ownership or influencing a team withou
How do you handle pressure?	Explain prioritization, communication, risk management and maintaining qu

STAR method: Situation → Task → Action → Result. Keep the Action section longest because it shows what you actually did.

LEADERSHIP PRINCIPLES PREPARATION

Principle	Prepare an example about
Customer Obsession	Improving a user/customer outcome
Ownership	Taking responsibility beyond your assigned task
Invent and Simplify	Removing unnecessary complexity
Learn and Be Curious	Learning a technology or solving an unfamiliar problem
Insist on Highest Standards	Catching quality issues and improving standards
Bias for Action	Making a sensible decision with incomplete information
Earn Trust	Building credibility through communication and reliable execution
Dive Deep	Finding the root cause of a difficult issue
Deliver Results	Completing an important goal with measurable impact

Preparation method: Prepare 6–8 real stories that can be adapted to multiple principles. Do not memorize a script word-for-word.

KEY LEARNINGS FROM THE EXPERIENCE

1. Start preparation early rather than trying to cover DSA, design and behavioral questions in the final week.
2. Practice coding without an IDE sometimes. This improves syntax recall, communication and debugging discipline.
3. Always state time and space complexity after solving a coding problem.
4. Practice follow-up questions because interviewers often change constraints after the first solution.
5. Be honest when you do not know something. Explain what you do know and how you would investigate the missing part.
6. Use your own projects as behavioral examples whenever possible.
7. Research the role and understand what the team actually builds before the interview.
8. Treat every round as an evaluation of communication as well as technical correctness.
9. Review mistakes after every mock interview and maintain a list of weak topics.
10. Stay calm when a problem feels unfamiliar. Clarify it, break it down and start with a valid baseline.

Core lesson: Strong candidates do not necessarily know every question beforehand. They demonstrate a repeatable process for understanding unfamiliar problems and reaching a defensible solution.

4-WEEK AMAZON-STYLE PREPARATION PLAN

Week	Focus	Daily target
Week 1	Arrays, strings, hashing, linked lists	2–3 DSA problems + 30 min CS fundamentals
Week 2	Trees, graphs, heaps, binary search	2–3 DSA problems + 1 timed set
Week 3	DP, sliding window, LRU + system design	2 DSA problems + 1 design discussion
Week 4	Behavioral, leadership stories, mocks	1 timed coding round + 1 behavioral mock

Final checklist: DSA fundamentals ✓ • coding under time pressure ✓ • complexity analysis ✓ • basic system design ✓ • 6–8 STAR stories ✓ • company/role research ✓ • questions for interviewer ✓